

Traveling Salesman Problem (TSP)

Model Documentation

Assignment-01

15-June-2026

1. Problem Definition

- From a set of cities (nodes), we need to route a vehicle to visit every single city exactly once.
- We take into account the real-world road network distances between every pair of cities.
- Our overall objective is to minimize the total travel distance (in kilometers) while ensuring the vehicle forms a single, continuous route that returns to its starting point.

INPUTS	TRAVELING SALESMAN OPTIMIZATION	OUTPUTS
City Coordinates →		→ Optimal sequence of cities (Tour)
Driving Distances →		→ Objective Bound & MIP Gap
Depot Location →		→ HTML map of final tour
Instance Size →		→ Total minimal driving distance

OBJECTIVE FUNCTION:

- Find the best route by minimising the total travel distance (in kilometers)

CONSTRAINTS:

- Leave every city exactly once
- Enter every city exactly once
- Eliminate disconnected subtours to ensure the vehicle forms one continuous loop

2. Data

Sets and Indices

Set	Index	Description
Cities (Nodes)	V	Set of all cities, indexed by i or j , where 0 is the Depot.
Non-Depot Cities	V'	Set of all cities excluding the Depot ($i \in \{1, \dots, N - 1\}$).
Valid Paths (Arcs)	A	Set of all valid directed paths (i, j) where $i \neq j$.

Parameters

Parameter	Description
N	Total number of cities in the current problem instance.
$\text{dist}(i, j)$	Real-world driving distance (in kilometers) from city i to city j , sourced from OSRM data.

3. Mathematical Formulation

Variables

- $X_{i,j} = \{0, 1\}$, if the route travels directly from city i to city j , for all Cities i, j where $i \neq j$
- $U_i \in \mathbb{R}$, bounded between 1 and $N - 1$, indicating the sequence order of city i in the tour, for all Non-Depot Cities

Objective

- Minimize Traveling distance:
$$\sum_{\text{Cities}} \text{dist}(i, j) * X_{i,j}$$

Constraints

- Vehicle must depart from every city exactly once:
$$\sum_{\text{Cities}} X_{i,j} = 1, \text{ for each } i \text{ in Cities}$$
- Vehicle must arrive at every city exactly once:
$$\sum_{\text{Cities}} X_{i,j} = 1, \text{ for each } j \text{ in Cities}$$
- Eliminate disconnected subtours to ensure one continuous loop:
$$U_i - U_j + N * X_{i,j} \leq N - 1, \text{ for each } i, j \text{ in Non-Depot Cities}$$

4. Counting the Number of Variables

Variable / Constraint Type	Formula	Number for Instance ($N = 1000$)
Binary Variables ($X_{i,j}$)	$N \times (N - 1)$	999,000
Continuous Variables (U_i)	$N - 1$	999
Degree Constraints	$2N$	2,000
MTZ Constraints	$(N - 1) \times (N - 2)$	997,002

5. Computational Results

FINAL RESULTS TABLE						
N (cities)	Optimal / Best Distance (km)	Runtime	MIP Gap (%)	Solver Status	Notes	
10	6176.00 / 6176.00	0.08 sec	0.00%	Optimal	Map saved as TSP_Map_N10.html	
50	11745.00 / 11745.00	50.42 sec	0.00%	Optimal	Map saved as TSP_Map_N50.html	
100	16818.00 / 16794.00	7200.04 sec	0.14%	Time-limited	Map saved as TSP_Map_N100.html	
250	26468.00 / 24311.00	7200.16 sec	8.15%	Time-limited	Map saved as TSP_Map_N250.html	
500	36607.00 / 33599.01	7209.77 sec	8.22%	Time-limited	Map saved as TSP_Map_N500.html	
750	45673.00 / 40469.00	7202.08 sec	11.39%	Time-limited	Map saved as TSP_Map_N750.html	
1000	56077.00 / 45721.00	7216.03 sec	18.47%	Time-limited	Map saved as TSP_Map_N1000.html	

6. HTML map

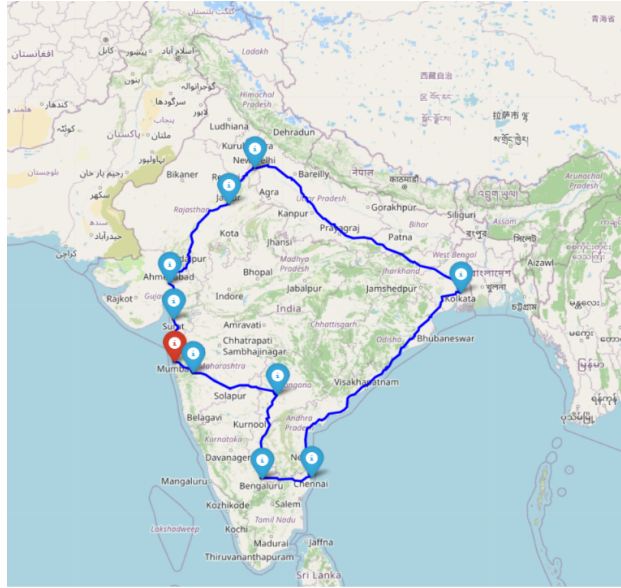


Figure 1: N=10 case

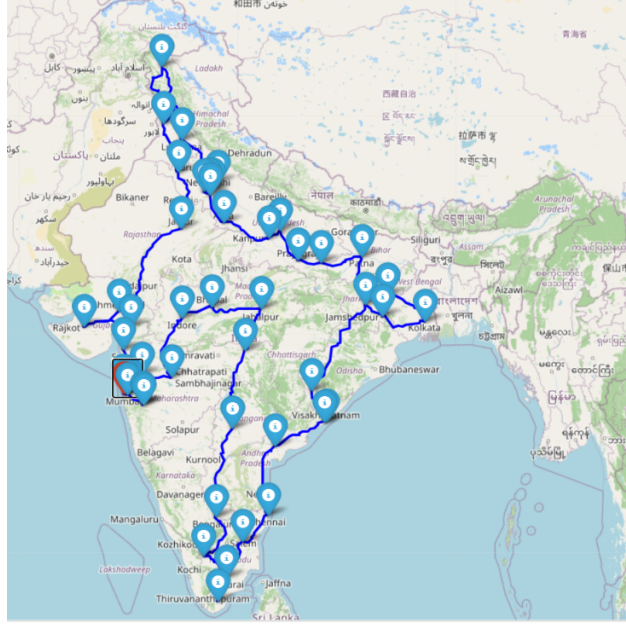


Figure 2: $N=50$ map

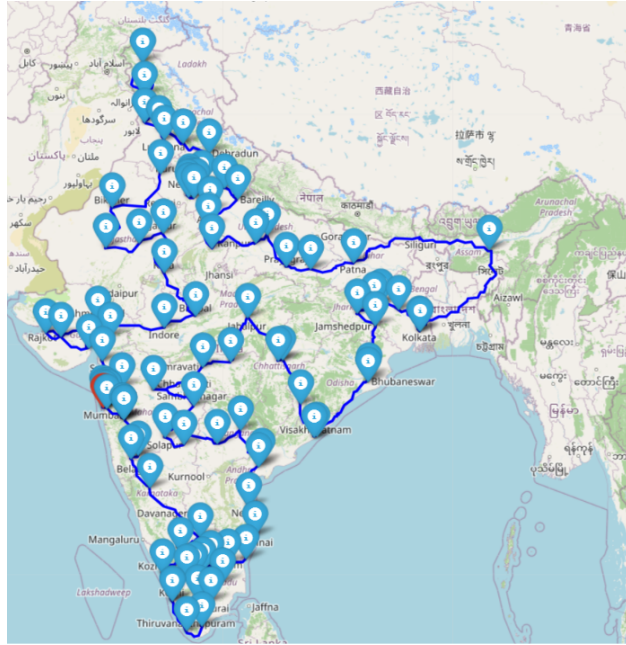


Figure 3: $N=100$ map

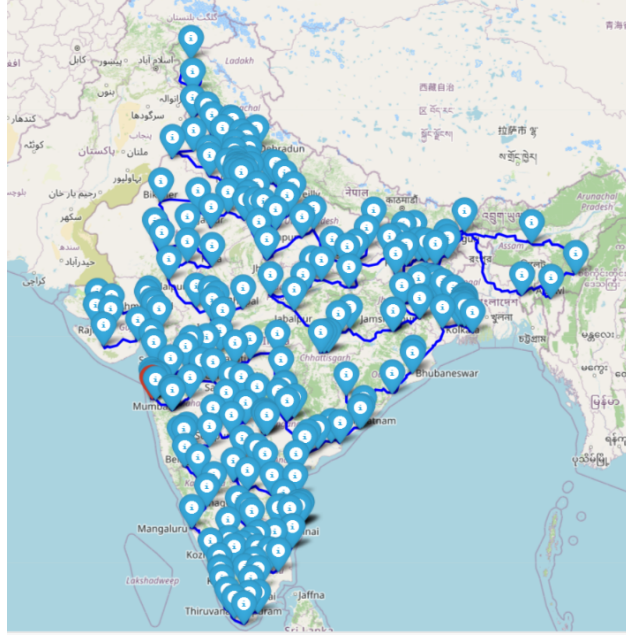


Figure 4: $N=250$ map

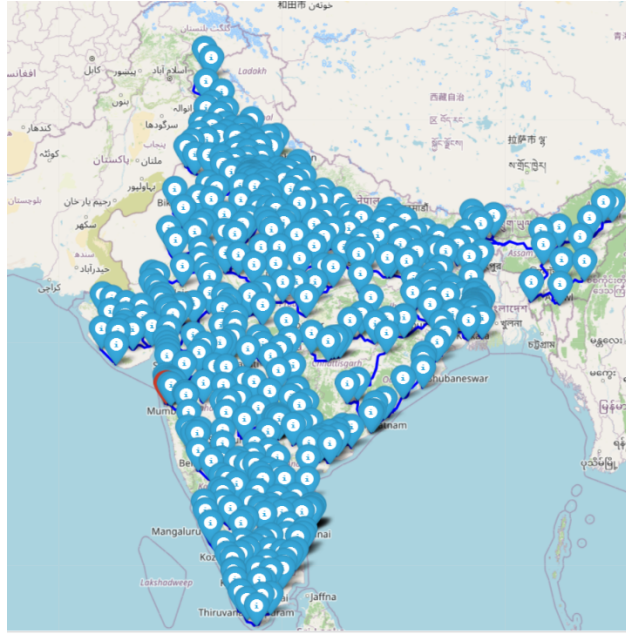


Figure 5: $N=500$ map

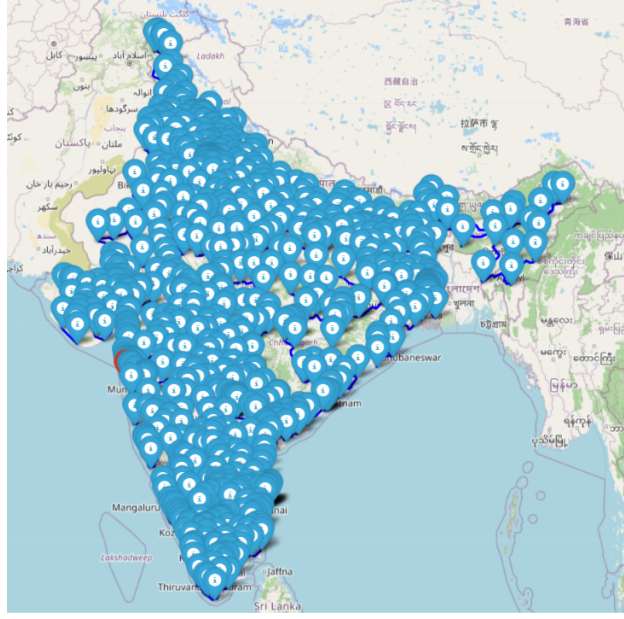


Figure 6: $N=750$ map

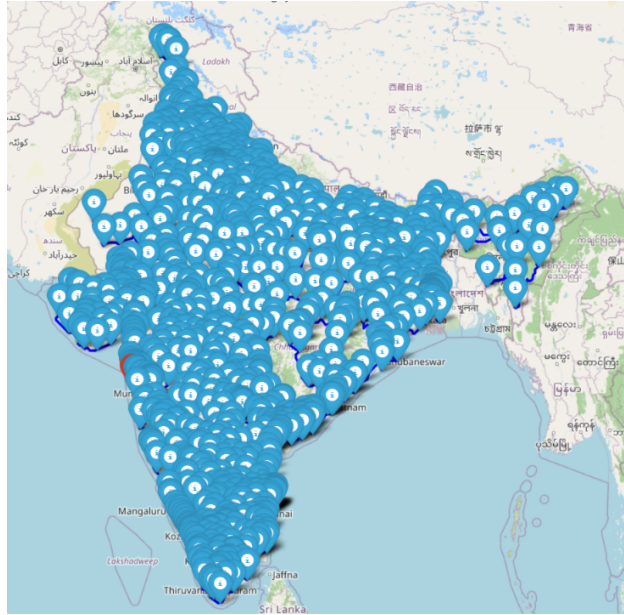


Figure 7: $N=1000$ map

6. Analysis

As the N value increases the gap and time taken are also increased. For N ($N \leq 50$) solver is able to get optimal solution under few seconds. But once N value goes to 100 the time taken has been increased exponentially. For $N = 750$ and $N = 1000$, the solver reaches the 7200-second limit before it can evaluate every branch.